

M1.5: Simple Harmonic Motion

1. Defining Equation

A particle undergoes **simple harmonic motion (SHM)** about the origin if its acceleration is proportional to displacement and directed towards the origin:

$$\ddot{x} = -\omega^2 x.$$

The positive constant ω is the **angular frequency**.

The general solution is

$$x(t) = A \cos(\omega t + \phi) = C \cos \omega t + D \sin \omega t$$

with amplitude A ($|x|$ max), phase ϕ , period $T = 2\pi/\omega$, and frequency $f = \omega/(2\pi)$.

2. Velocity and Energy Relations

$$v^2 = \omega^2(A^2 - x^2)$$

This follows by integration: max speed $v_{\max} = A\omega$ at $x = 0$, zero speed at $x = \pm A$.

Energy

$$\text{KE} = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(A^2 - x^2)$$

$$\text{PE} = \frac{1}{2}m\omega^2 x^2 \text{ (relative to equilibrium)}$$

$$\text{Total} = \frac{1}{2}m\omega^2 A^2 \text{ (constant).}$$

3. Springs (Hooke's Law)

A particle of mass m attached to a light spring of natural length ℓ_0 and stiffness k (force constant). At extension x from natural length, spring force = $-kx$ (restoring).

If the system is displaced from equilibrium by y , Newton's second law gives $m\ddot{y} = -ky$, so SHM with

$$\omega = \sqrt{k/m}, \quad T = 2\pi\sqrt{m/k}.$$

Spring Hanging Vertically

Equilibrium extension e satisfies $ke = mg$. Displacement from equilibrium y satisfies $m\ddot{y} = -ky$ — same SHM as horizontal. Gravity shifts the equilibrium but does not affect the period.

Elastic Strings (Modulus of Elasticity λ)

For a string of natural length ℓ_0 and modulus λ , the tension at extension x is $T = \lambda x/\ell_0$. The effective stiffness is $k = \lambda/\ell_0$.

Caveat: strings cannot be compressed — tension is zero when "extension" would be negative. SHM applies only while the string is taut.

A particle of mass 2 kg hangs from a spring of natural length 0.5 m and modulus 20 N extended to 1 m equilibrium. Find the period of small oscillations.

Stiffness: $k = \lambda/\ell_0 = 20/0.5 = 40 \text{ N/m}$.

$$\omega = \sqrt{40/2} = \sqrt{20}. \quad T = 2\pi/\omega = 2\pi/\sqrt{20} \approx 1.40 \text{ s}.$$

4. Simple Pendulum

For small-angle oscillations of a simple pendulum of length ℓ :

$$\ddot{\theta} = -\frac{g}{\ell} \sin \theta \approx -\frac{g}{\ell} \theta \quad (\text{for small } \theta)$$

SHM with $\omega = \sqrt{g/\ell}$, $T = 2\pi\sqrt{\ell/g}$. Period is independent of amplitude (for small swings) and mass.

5. Time, Position, Velocity

Given $x = A \cos(\omega t)$ with $x_0 = A$, $\dot{x}_0 = 0$:

- Time to first zero: $\omega t = \pi/2$, so $t = T/4$.
- Time from $x = A$ to $x = A/2$: $\cos \omega t = 1/2$, so $\omega t = \pi/3$, $t = T/6$.

If the particle starts at $x = 0$ moving in the positive direction with speed $v_0 = A\omega$, use $x = A \sin(\omega t)$ instead.

6. Combining Conditions

Questions often give you two data points (e.g. v at a specific x , or position at a specific t). Use:

- $v^2 = \omega^2(A^2 - x^2)$ to link v , x , ω , A .
- Initial conditions to fix the phase ϕ in $x(t) = A \cos(\omega t + \phi)$.

Particle executes SHM with period 4 s and amplitude 5 cm. Find its speed when $x = 3$ cm.

$$\omega = 2\pi/4 = \pi/2 \text{ rad/s. } v^2 = (\pi/2)^2(5^2 - 3^2) = (\pi^2/4) \cdot 16 = 4\pi^2.$$

$$|v| = 2\pi \approx 6.28 \text{ cm/s.}$$

7. Damped SHM (Contextual)

Real springs have damping. A linear damping force $-c\dot{x}$ gives

$$m\ddot{x} + c\dot{x} + kx = 0.$$

This is a second-order DE (see §1.8) and exhibits under-, critical-, or over-damping depending on c^2 vs. $4mk$. Only the undamped case is classical SHM.

8. Exam Style Strategy

- Always identify ω first — from the differential equation or from T .
- Choose an appropriate parametrisation (cos if starting at max displacement, sin if starting at equilibrium moving).
- Use energy / $v^2 = \omega^2(A^2 - x^2)$ for "speed at this x " questions.
- For elastic strings, check whether the string is taut throughout — SHM may break down if the particle overshoots.